Functions

Tuesday, September 29, 2026 6:40 PM

Straight Line

$y = mx + c$ / $y - y_1 = m(x - x_1)$

m – gradient

c – y-intercept

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$m > 0$, the graph is increasing

$m < 0$, the graph is decreasing

$m = 5 > 0$

$m = -6 < 0$

Domain: $x \in \mathbb{R}$

Range: $y \in \mathbb{R}$

Parabola

$y = f(x) = a(x \pm p)^2 \pm q = ax^2 + bx + c$

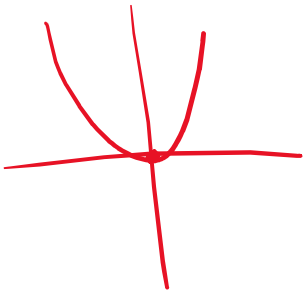
$TP(p; q)$ — axis of symmetry.

TP: $x = -\frac{b}{2a}$; $y = f(-\frac{b}{2a})$

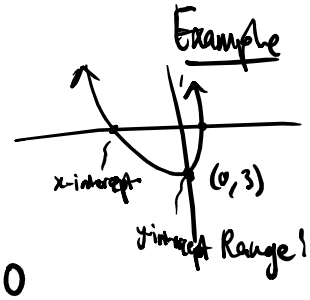
$a < 0$: concave down ∩

$a > 0$: concave up ∪

y-intercept



Example



Range: $y \in [3, \infty)$

Domain: $x \in \mathbb{R}$

Range: $y \in [q; \infty)$ / $y \geq q$ if $a > 0$

$y \in (-\infty, q]$ / $y \leq q$ if $a < 0$

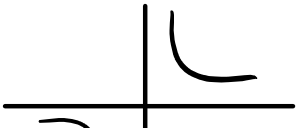
Hyperbola

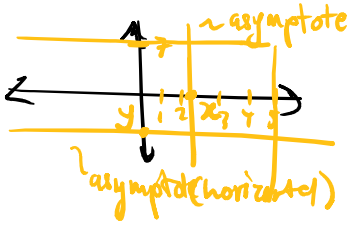
$y = \frac{a}{x \pm p} \pm q$ ← $x + 2 = 0$, $x = -2$

Vertical asymptote: $x = p$

Horizontal asymptote: $y = q$

$a > 0$:





$y = \frac{a}{x}$

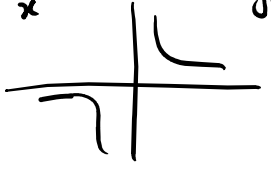
example

$f(x) = \frac{2}{x-5} + 7$

Vertical asymptote: $x - 5 = 0$, $x = 5$

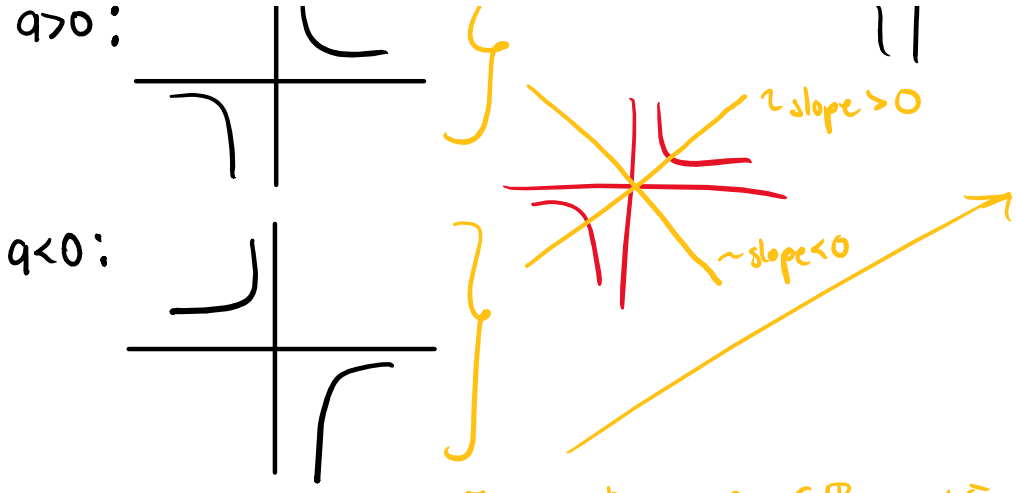
Horizontal asymptote: $y = 7$

~Hyperbola



$q>0$:

$q<0$:



Example: Domain: $x \in \mathbb{R}, x \neq 5$

Example, Range: $y \in \mathbb{R}, y \neq 7$

Domain: $x \in \mathbb{R}, x \neq \pm p$

Range: $y \in \mathbb{R}, y \neq q$

Axis of symmetry

$y = \pm(x+p)+q$

$\begin{cases} y = +(x+p)+q \\ y = -(x+p)+q \end{cases}$

Exponential

$y = f(x)$

$y = a \cdot b^{x \pm p} \pm q$

$1^{x+5} = 1$

$b > 0$ and $b \neq 1$



Horizontal asymptote: $y = q$

$a > 0$: graph is above the asymptote.

$a < 0$: graph is below the asymptote.

Domain: $x \in \mathbb{R}$

Range: $y \in (q; \infty) / y > q$ if $a > 0$

$y \in (-\infty, q) / y < q$ if $a < 0$

Examples

1. A straight-line graph passes through two points, namely $(-2; -3)$ and $(5; 7)$.

(a) Determine the equation of the graph.

$m = \frac{y_2 - y_1}{} \quad \frac{7-(-3)}{} \quad$ in

Domain: $x \in \mathbb{R}$

(a) Determine the equation of the graph.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7 - (-3)}{5 - (-2)} = \frac{10}{7}$$

Domain: $x \in \mathbb{R}$
Range: $y \in \mathbb{R}$

$$y - y_1 = m(x - x_1)$$
$$y - (-3) = \frac{10}{7}(x - (-2))$$
$$y + 3 = \frac{10}{7}(x + 2)$$
$$y = \frac{10}{7}x + \frac{20}{7} - 3$$
$$y = \frac{10}{7}x - \frac{1}{7}$$

x-intercept: $y = 0$

$$\frac{10}{7}x - \frac{1}{7} = 0$$
$$\left[\frac{10}{7}x = \frac{1}{7}\right] \times \frac{7}{10}$$
$$x = \frac{1}{10}$$

2. Given: $f(x) = -2x^2 + x + 6$

Domain: $x \in \mathbb{R}$
Range: $y \in (-\infty, \frac{49}{8})$

(a) Determine the y-intercept. $y = 6$

(b) Determine the x-intercepts. $y = 0$

$$-2x^2 + x + 6 = 0$$
$$(-2x - 3)(x - 2) = 0$$
$$x = -\frac{3}{2} \text{ or } x = 2$$

(c) Determine the coordinates of the TP. $x = \frac{-b}{2a} = \frac{-1}{2(-2)} = \frac{1}{4}$

$$f(\tfrac{1}{4}) = -2(\tfrac{1}{4})^2 + \tfrac{1}{4} + 6 = \frac{49}{8}$$

$TP(\frac{1}{4}; \frac{49}{8})$

3. Given the equation of the hyperbola: $f(x) = \frac{-4}{2 - x} - 1$

Domain: $x \in \mathbb{R}, x \neq 2$
Range: $y \in \mathbb{R}, y \neq -1$

(a) Write down the equations of the asymptotes of f.

Vertical asymptote: $2 - x = 0$
$x = 2$

Vertical asymptote: $2-x=0$, $x=2$

Horizontal asymptote: $y=-1$

(b) Determine the intercepts of the graph of f with the axis.

y-intercept: $y=f(0)=\frac{-4}{2-0}-1=-3$

x-intercept: $0=\frac{-4}{2-x}-1$

$1=\frac{-4}{2-x}$

$2-x=-4$

$-x=-6$

$x=6$

4. Given the equation of the graph: $f(x)=\frac{x+6}{x+2}$

(a) Express $f(x)$ in the form of $f(x)=\frac{a}{x+p}+q$

$f(x)=\frac{x+6}{x+2}=\frac{x+2+4}{x+2}=\frac{x+2}{x+2}\frac{1}{}+\frac{4}{x+2}=\frac{4}{x+2}+1$

Vertical asymptote: $x=-2$

Horizontal asymptote: $y=1$

Domain: $x\in\mathbb{R}, x\neq -2$

Range: $y\in\mathbb{R}, y\neq 1$

Axis of symmetry:

$y=\frac{4}{x+2}+1$

$y=(x+2)+1=x+3$ ~ y-intercept

$y=-(x+2)+1=-x-1$ ~ y-intercept



Axis of symmetry (Parabola example)

$f(x)=2x^2+5x+1$

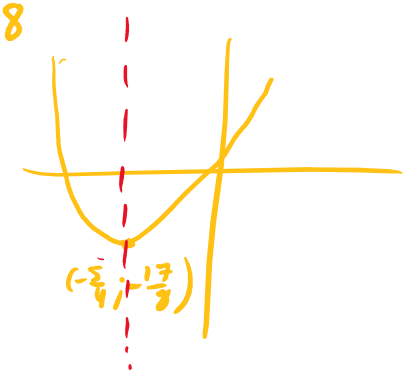
$x=\frac{-b}{2a}=\frac{-5}{2(2)}=-\frac{5}{4}$

$y=f(-\frac{5}{4})=2(-\frac{5}{4})^2+5(-\frac{5}{4})+1=-\frac{17}{8}$

$y = f(-\frac{5}{4}) = 2(-\frac{5}{4})^2 + 5(-\frac{5}{4}) + 1 = -\frac{17}{8}$

TP $(-\frac{5}{4}; -\frac{17}{8})$

Axis of symmetry: $x = -\frac{5}{4}$



Nature of roots

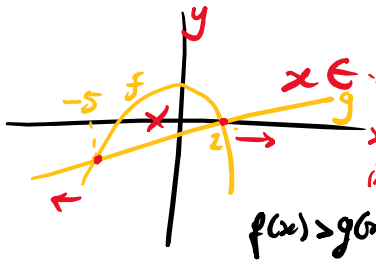
$\Delta = b^2 - 4ac$, descriminant.

- $b^2 - 4ac > 0$ ~ real and unequal roots
- $b^2 - 4ac > 0$ and a perfect square ~ real, unequal and rational roots
- $b^2 - 4ac > 0$ and not a perfect square ~ real, unequal and irrational roots
- $b^2 - 4ac = 0$ ~ real and equal roots
- $b^2 - 4ac < 0$ ~ non-real roots

example: $b^2 - 4ac = -1$ $\sqrt{-1}$

Things to know

- $f(x) = g(x)$
- $f(x) < g(x)$ / $f(x) - g(x) < 0$
- $f(x) > g(x)$ / $f(x) - g(x) > 0$
- $f(x) \cdot g(x) > 0$ / $f(x) \cdot g(x) < 0$
- $\frac{f(x)}{g(x)} > 0$



$f(x) < g(x)$

$x \in$ (−∞, 5) or (2, ∞)

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$f(x) > g(x)$: $x \in (-5; 2)$

Examples

1. Given: $f(x) = -2x^2 + x + 6$

1.1 Determine the value(s) of k for which $f(x) = k$ will have equal roots.

$f(x) = k$

$-2x^2 + x + 6 = k$

$g(x) = k$

$-2x^2 + x + 6 = k$

$-2x^2 + x + (6-k) = 0$

$* a = -2 ; b = 1 ; c = 6-k$

$\Delta = b^2 - 4ac = 0$

$(1)^2 - 4(-2)(6-k) = 0$

$1 + 8(6-k) = 0$

$\frac{8(6-k)}{8} = \frac{-1}{8}$

$6 - k = -\frac{1}{8}$

$-k = -\frac{1}{8} - 6$

$-k = -\frac{49}{8}$

$k = \frac{49}{8}$

1.2 If the graph is shifted 2 units to the right and 1 unit upwards to form the graph of h, determine the equation of h in the form of $y = a(x+p)^2 + q$

$TP\left(\frac{1}{4}, \frac{49}{8}\right)$

$f(x) = -2\left(x - \frac{1}{4}\right)^2 + \frac{49}{8}$

$h(x) = -2\left(x - \frac{9}{4}\right)^2 + \frac{57}{8}$

2. Given: $h(x) = 4(2^x) + 1$

2.1 Determine the coordinates of the y-intercept

$h(0) = 4(2^0) + 1 = 4(1) + 1 = 5$

Coordinates: $(0; 5)$

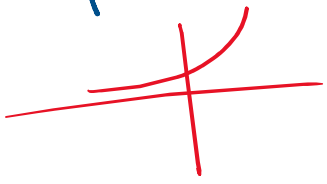
2.2 Explain why h does not have an x-intercept.

$h(x) = 0$

$4(2^x) + 1 = 0$

$4(2^x) = -1$

$2^x = -\frac{1}{4}$ N/S



$2^x = -\frac{1}{4}$ N/S

$2^x > 0$ for all $\mathbb{R}$ x

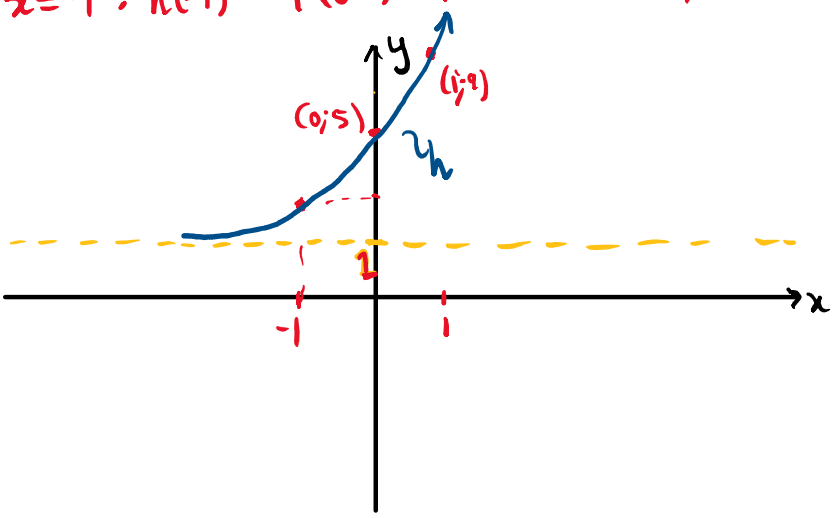
2.3 Sketch the graph h, showing all asymptote(s), intercept with the axes and atleast one other point on h.

Horizontal asymptote : $y = 1$

y-intercept: $(0;5)$

At $x=1$: $h(1) = 4(2^1)+1 = 9$ $\therefore (1;9)$

At $x=-1$: $h(-1) = 4(2^{-1})+1 = 3$ $\therefore (-1;3)$



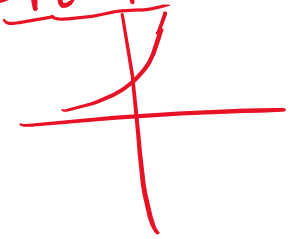
2.4 Describe the transformation from h to g if $g(x) = 4(2^{-x}+2) = 4\cdot 2^{-x}+8$

$h(-x) = 4(2^{-x})+1$

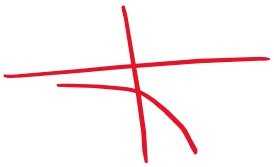
$h(-x) = 4(2^{-x})+1+7 = 4(2^{-x})+8 = g(x)$

$\therefore$ There's a reflection about the y-axis and a vertical shift of 7 units upwards

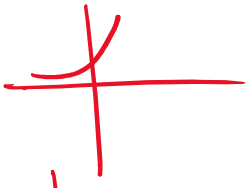
$y = a^x$:



$y = a^{-x}$:



$y = a^x$:



$y = -a^x$




y = -a^x
1